CLDV 6211 Formative

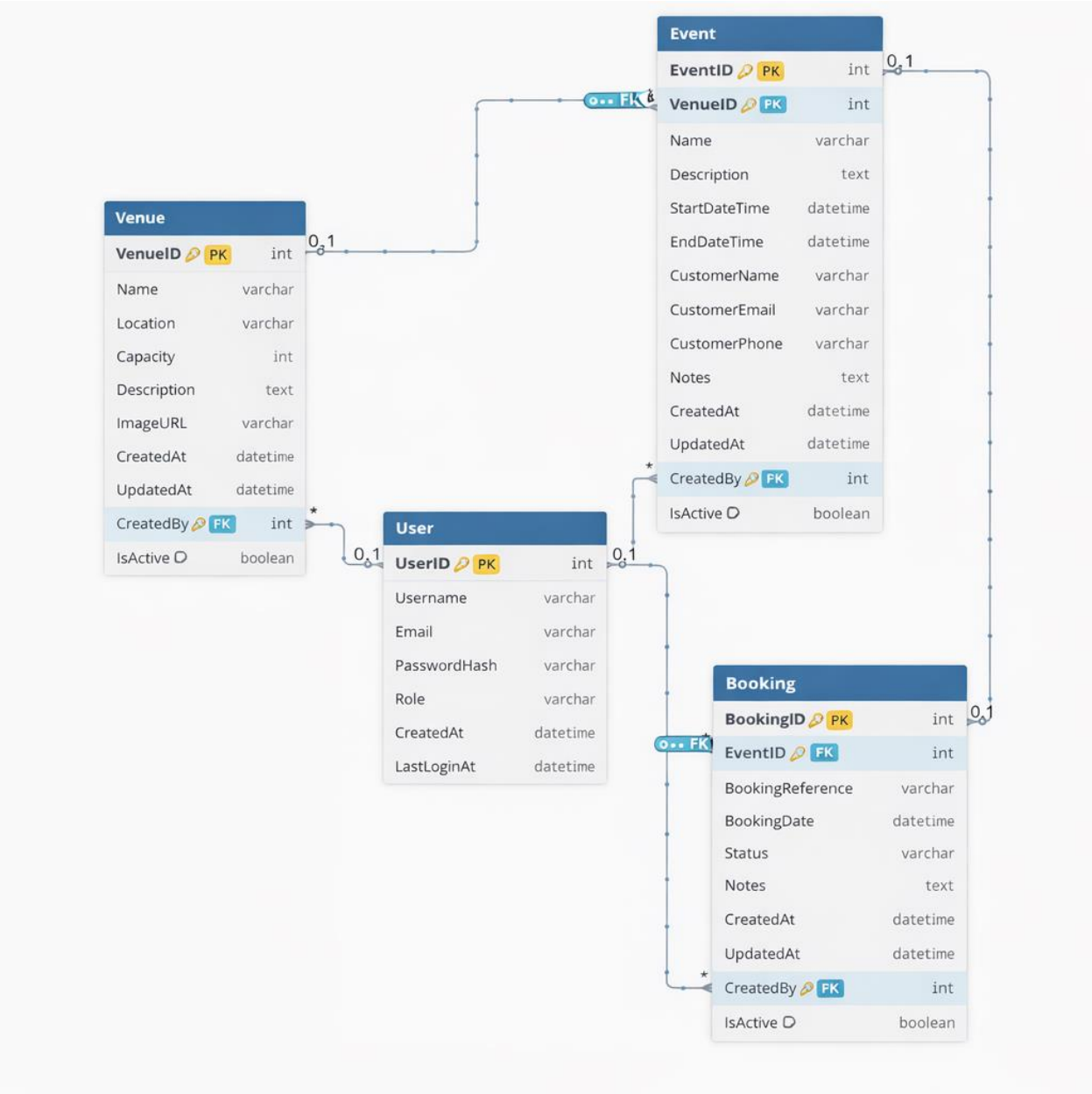
https://github.com/VanNiekerkJoe/EventEase_ST10445055_Group3_POE-.git

https://www.youtube.com/watch?v=zZV7uR6Nq4I

ST10445055

Group 3

Formative 1 Part 1



Cloud Computing Basics

1. In what ways does deploying an app in the cloud differ from running it on-premises, especially around security, deployment speed, and resource management?

- **Security** On-premises means your team controls literally everything—the firewalls, who can walk into the server room, encryption setup, patching schedules. If you're in a regulated industry, that level of hands-on is sometimes required to prove compliance. But it also means every little thing is your problem. Forget to update something? You're the one dealing with it. Cloud providers handle a ton of the foundational stuff: physical data center security, massive DDoS protection, automatic patching at scale, encryption options that are on by default for most things. You still own your app's logic, access controls, and data protection rules, but they catch a lot of common attacks faster because they're seeing patterns across huge numbers of users. Like, a new exploit shows up—cloud side often blocks it globally way quicker than one team patching servers manually.

- **Deployment speed** On-prem is slow. You have to spec hardware, wait for procurement, rack it, install OS, configure networks, test failover… even a tiny update can mean coordinating downtime. Weeks or months to get something new live isn't unusual. Cloud? You click a few things or run a command, and a server (or whole stack) is ready in minutes. Templates and pipelines make pushing code feel almost instant. For EventEase, if someone wants to try out a new venue filter or availability view, you can have a test version running that afternoon—no waiting on budget approvals or hardware shipments.

- **Resource management** With on-prem, you guess your peak needs and buy enough servers to cover it. Wedding season hits hard? Hope it holds. Off-season? Those machines sit mostly idle, eating electricity and depreciation. Scaling means more purchases and setup time. Cloud billing is usage-based—you pay for what runs. Traffic spikes from a popular venue listing? It adds capacity automatically (or close to it). Slow

period? Costs drop without you touching anything. No more buying "just in case" hardware that collects dust 10 months out of the year.

Cloud isn't perfect—you trade some direct control—but you get faster everything, easier ups and downs, and usually better day-to-day protection. For a company growing like EventEase, it's hard to beat when you don't want to become a full-time server wrangler.

2. What are the key differences between IaaS, PaaS, and SaaS? Why might EventEase benefit from PaaS over the others when they eventually go cloud?

- **IaaS (Infrastructure as a Service)** You're renting virtual machines, storage, and basic networking. Total freedom—you install the OS, pick your database, set up web servers, handle patching and scaling scripts yourself. It's like getting a blank server rack in someone else's building. You control everything below your app, but you still do all the sysadmin work. AWS EC2, Azure VMs, Google Compute—classic examples. Fine if you need weird custom setups or old software that won't run anywhere else.

- **PaaS (Platform as a Service)** The provider runs the servers, keeps the OS updated, manages databases, handles load balancing, auto-scales when traffic jumps. You write your code, deploy it (often just git push or a button), connect your data, and that's mostly it. Less setup, less ongoing hassle. Heroku, AWS Elastic Beanstalk, Google App Engine, Azure App Service—those kinds of things. Built for teams who want to ship features instead of babysit infrastructure.

- **SaaS (Software as a Service)** Complete app you just sign into. Provider does every layer—code, servers, updates, scaling. Gmail, Salesforce, Calendly, Zoom. You configure users and settings, maybe add some integrations, but no real development or server stuff. It’s the easiest option even if the user is limited to a lot of pre existing features.

For EventEase's setup—a custom admin platform with venue browsing, event scheduling, double-booking prevention logic, image uploads for venues, and the specific flow where specialists take phone/email requests and enter bookings—**PaaS looks like the practical choice**.

SaaS is out because off-the-shelf tools don't let you build in those exact rules (like checking venue dates for overlaps) or tweak the workflow for admin-only use. You'd be forcing your process to fit their product, not the other way around.

IaaS could technically run it, sure. But then the team ends up spending half their time on VM sizing, OS patches, monitoring alerts, writing auto-scaling rules—stuff that has nothing to do with making the booking system better. That's time not spent on things like nicer search filters or email notifications for confirmations.

With PaaS, they keep writing their own code and database queries (the custom bits that make it EventEase), but skip most of the grunt work below that. When bookings ramp up (seasonal peaks or a hot venue), it scales without manual intervention. Deploying a small fix? Quick and painless. Venue photos get stored securely through built-in file handling. Costs stay reasonable because you only pay for what's actually used—no idle servers.

For a smaller/growing team planning to scale this thing over time (maybe add customer self-service later), PaaS usually means faster progress, lower ops overhead, and money spent on features instead of infrastructure. That's probably why it fits their roadmap best when they make the cloud jump.

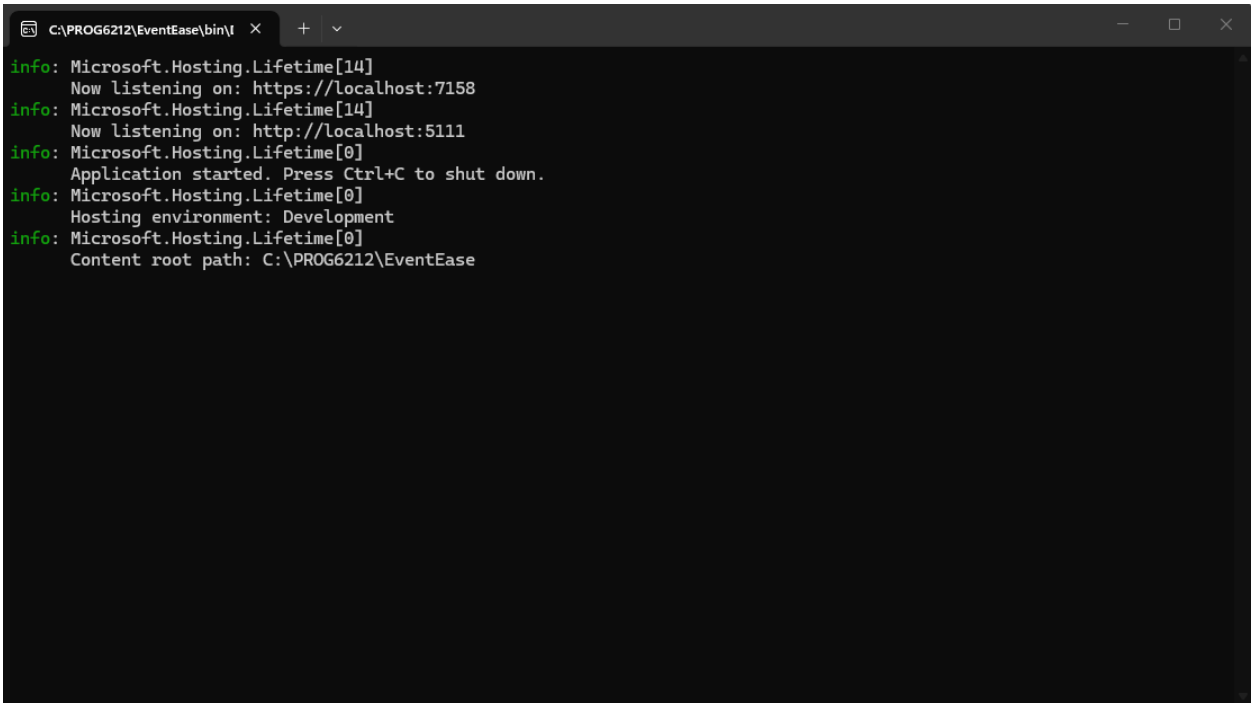
Reference list

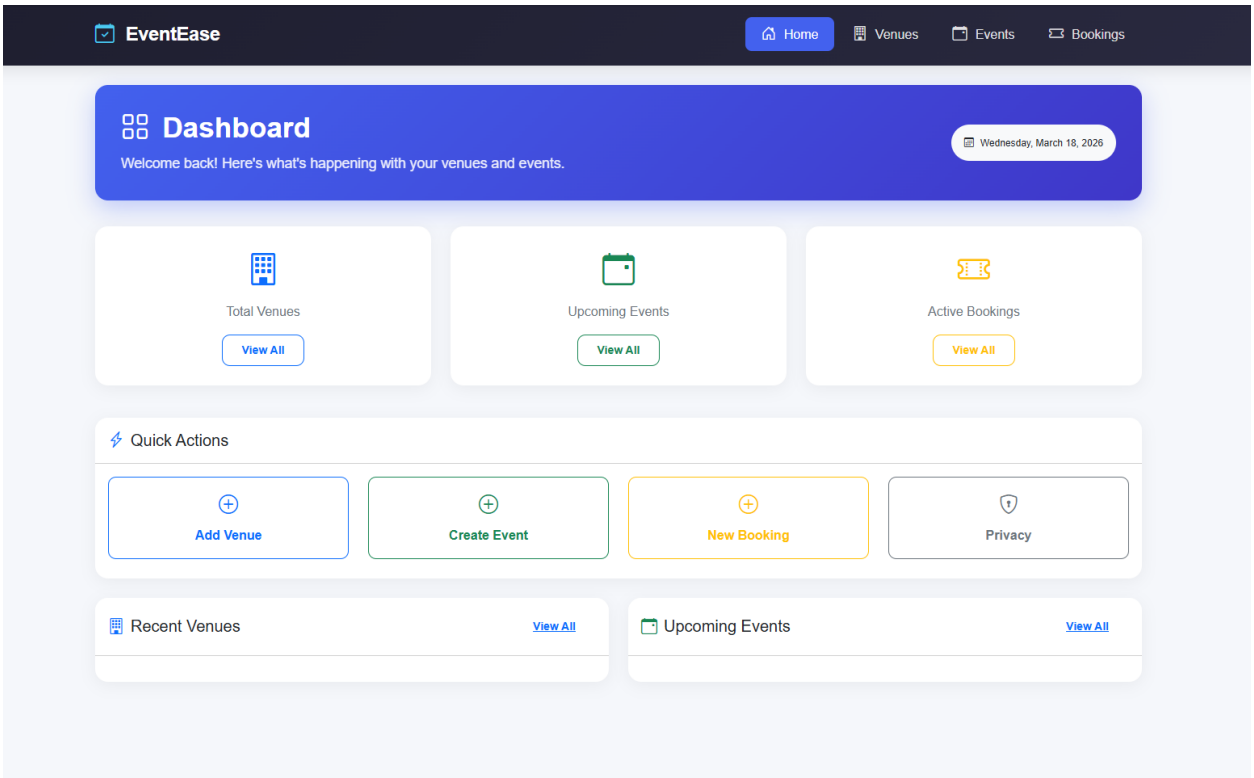
Dataconomy. (2025). *Cloud hosting vs local servers: Pros, cons and costs*. [online] Available at: https://dataconomy.com/2025/04/08/cloud-hosting-vs-local-servers-pros-cons-and-costs/.

Google Cloud (n.d.). *PaaS vs IaaS vs SaaS: What's the difference?* [online] Google Cloud. Available at: https://cloud.google.com/learn/paas-vs-iaas-vs-saas.

LucidChart (2024). *What is an Entity Relationship Diagram (ERD)?* [online] Lucidchart. Available at: https://www.lucidchart.com/pages/er-diagrams.

Photos of running application :





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EventEase

Home | Venues | Events | Bookings

Venues

Create New Venue

Search by name or location...

Search | Clear

Grand Conference Hall

Grand Conference Hall
Downtown
Capacity: 500

Edit | Details | Delete

Garden Pavilion

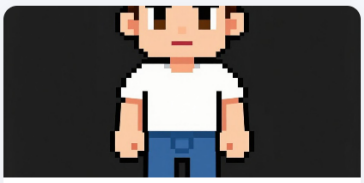
Garden Pavilion
North Side
Capacity: 200

Edit | Details | Delete

Executive Boardroom

Executive Boardroom
Business District
Capacity: 50

Edit | Details | Delete



Leo's house
Ruimsig
Capacity: 40

Edit | Details | Delete

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EventEase

Home | Venues | Events | Bookings

Edit Venue

Venue Name

Leo's house

Location

Ruimsig

Capacity

40

Current Image

Upload New Image

Choose file | No file chosen

Upload a new image to replace the current one

Back to List | Save Changes

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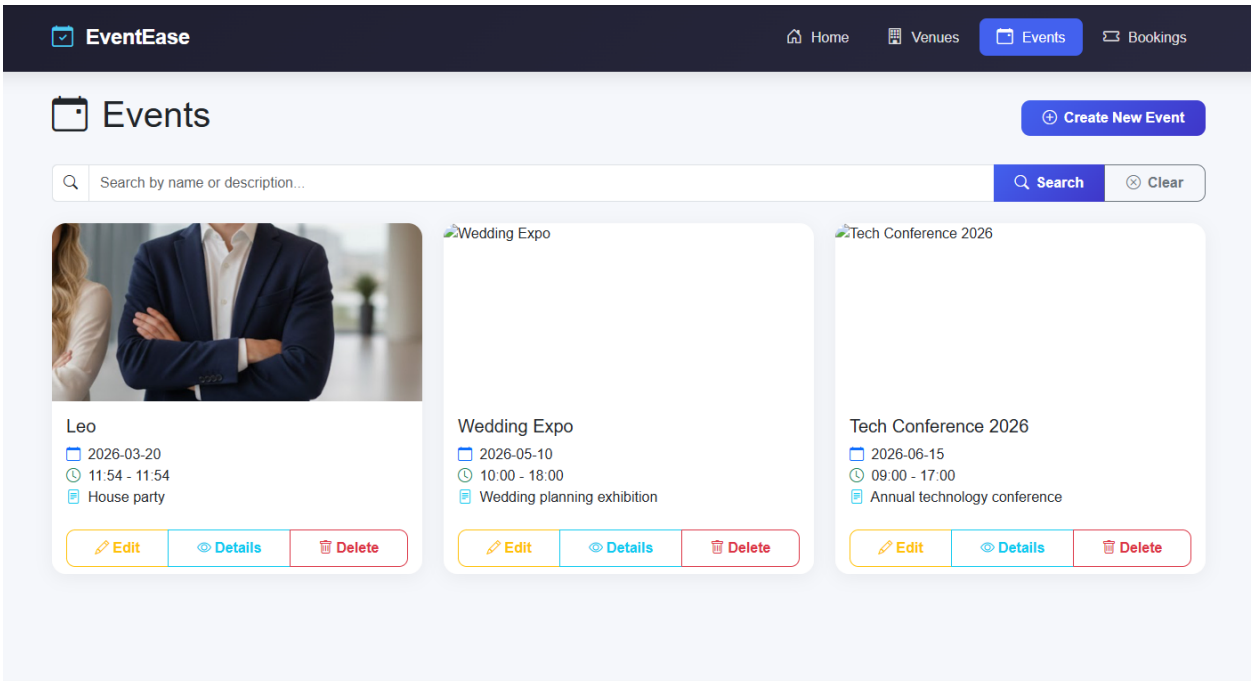

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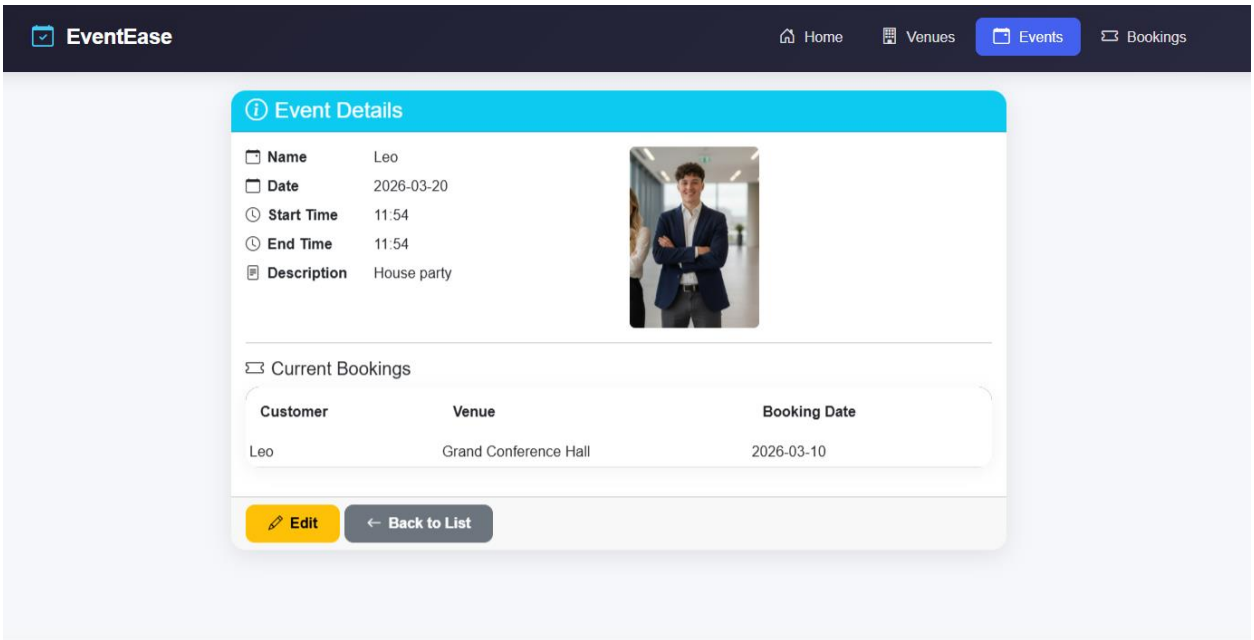

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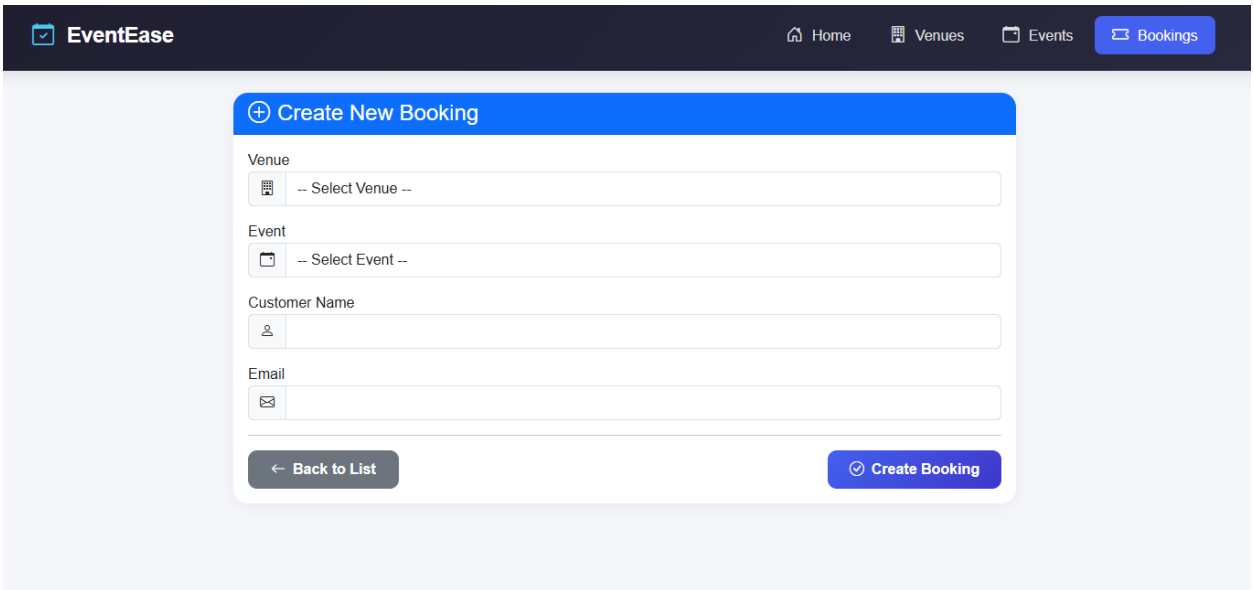

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